ASSEMBLY 8085

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OPCODE 8085

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00011010101001010010101

Μ.Ι.Σ. 2021-2022

Ο ASSEMBLER του 8085 δέχεται τις εντολές ASSEMBLY τις αναγνωρίζει σαν OPCODES και τα μετατρέπει σε 0 και 1.

το πρώτο byte είναι το OpCode της εντολής

Opcode	Operand
MVI	A, 32H
MVI	B, F2H

σε δυαδικό	δεκαξαδικό
0011 1110	3EH
0011 0010	32H
0000 0110	06H
1111 0010	F2H

το δεύτερο byte είναι ο operand της εντολής

OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC
00	NOP	2B	DCX H	56	MOV D,M	81	ADD C	AC	XRA H	D7	RST 2
01	LXI B,D16	2C	INR L	57	MOV D,A	82	ADD D	AD	XRA L	D8	RC
02	STAX B	2D	DCR L	58	MOV E,B	83	ADD E	AE	XRA M	D9	—
03	INX B	2E	MVI L,D8	59	MOV E,C	84	ADD H	AF	XRA A	DA	JC Adr
04	INR B	2F	CMA	5A	MOV E,D	85	ADD L	B0	ORA B	DB	IN D8
05	DCR B	30	SIM	5B	MOV E,E	86	ADD M	B1	ORA C	DC	CC Adr
06	MVI B,D8	31	LXI SPD16	5C	MOV E,H	87	ADD A	B2	ORA D	DD	—
07	RLC	32	STA Adr	5D	MOV E,L	88	ADC B	B3	ORA E	DE	SBI D8
08	—	33	INX SP	5E	MOV E,M	89	ADC C	B4	ORA H	DF	RST 3
09	DAD B	34	INR M	5F	MOV E,A	8A	ADC D	B5	ORA L	E0	RPO
0A	LDAXB	35	DCR M	60	MOV H,B	8B	ADC E	B6	ORA M	E1	POP H
0B	DCX B	36	MVI M,D8	61	MOV H,C	8C	ADC H	B7	ORA A	E2	JPO Adr
0C	INR C	37	STC	62	MOV H,D	8D	ADC L	B8	CMP B	E3	XTHL
0D	DCR C	38	—	63	MOV H,E	8E	ADC M	B9	CMP C	E4	CPO Adr
0E	MVI C,D8	39	DAD SP	64	MOV H,H	8F	ADC A	BA	CMP D	E5	PUSH H
0F	RRC	3A	LDA Adr	65	MOV H,L	8G	SUB B	BB	CMP E	E6	ANI D8
10	—	3B	DCX SP	66	MOV H,M	91	SUB C	BC	CMP H	E7	RST 4
11	LXI D,D16	3C	INR A	67	MOV H,A	92	SUB D	BD	CMP L	E8	RPE
12	STAX D	3D	DCR A	68	MOV L,B	93	SUB E	BE	CMP M	E9	PCHL
13	INX D	3E	MVI A,D8	69	MOV L,C	94	SUB H	BF	CMP A	EA	JPE Adr
14	INR D	3F	CMC	6A	MOV L,D	95	SUB L	C0	RNZ	EB	XCHG
15	DCR D	40	MOV B,B	6B	MOV L,E	96	SUB M	C1	POP B	EC	CPE Adr

OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC
16	MVI D,D8	41	MOV B,C	6C	MOV L,H
17	RAL	42	MOV B,D	6D	MOV L,L
18	—	43	MOV B,E	6E	MOV L,M
19	DAD D	44	MOV B,H	6F	MOV L,A
1A	LDAXD	45	MOV B,L	70	MOV M,B
1B	DCX D	46	MOV B,M	71	MOV M,C
1C	INR E	47	MOV B,A	72	MOV M,D
1D	DRC E	48	MOV C,B	73	MOV M,E
1E	MVI E,D8	49	MOV C,C	74	MOV M,H
1F	RAR	4A	MOV C,D	75	MOV M,L
20	RIM	4B	MOV C,E	76	HLT
21	LXI H,D16	4C	MOV C,H	77	MOV M,A
22	SHLD Adr	4D	MOV C,L	78	MOV A,B
23	INX H	4E	MOV C,M	79	MOV A,C
24	INR H	4F	MOV C,A	7A	MOV A,D
25	DCR H	50	MOV D,B	7B	MOV A,E
26	MVI H,D8	51	MOV D,C	7C	MOV A,H
27	DAA	52	MOV D,D	7D	MOV A,L
28	—	53	MOV D,E	7E	MOV A,M
29	DAD H	54	MOV D,H	7F	MOV A,A
2A	LHLD Adr	55	MOV D,L	80	ADD B

OP CODE	MNEMONIC	OP CODE	MNEMONIC	OP CODE	MNEMONIC
97	SUB A	C2	JNZ Adr	ED	—
98	SBB B	C3	JMP Adr	EE	XRI D8
99	SBB C	C4	CNZ Adr	EF	RST 5
9A	SBB D	C5	PUSH B	F0	RP
9B	SBB E	C6	ADI D8	F1	POP PSW
9C	SBB H	C7	RST 0	F2	JP Adr
9D	SBB L	C8	RZ	F3	DI
9E	SBB M	C9	RET Adr	F4	CP Adr
9F	SBB A	CA	JZ	F5	PUSH PSW
A0	ANA B	CB	—	F6	ORI D8
A1	ANA C	CC	CZ Adr	F7	RST 6
A2	ANA D	CD	CALL Adr	F8	RM
A3	ANA E	CE	ACI D8	F9	SPHL
A4	ANA H	CF	RST 1	FA	JM Adr
A5	ANA L	D0	RNC	FB	EI
A6	ANA M	D1	POP D	FC	CM Adr
A7	ANA A	D2	JNC Adr	FD	—
A8	XRA B	D3	OUT D8	FE	CPI D8
A9	XRA C	D4	CNC Adr	FF	RST 7
AA	XRA D	D5	PUSH D		
AB	XRA E	D6	SUI D8		

D8 = constant, or logical/arithmetic expression that evaluates to an 8 bit data quantity.

Adr = 16-bit address

D16 = constant, or logical/arithmetic expression that evaluates to a 16 bit data quantity